

KSHITIJ AGARWAL

Data Scientist | M.Sc.

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SUMMARY

I have been working in data science for over **10 years**. With an engineering background, I have a particular interest in understanding the world through probabilistic and statistical measurements. I have developed a mathematical model of how the **human brain functions** to keep us sitting upright, built a real-time algorithm that operates 24/7 to **detect and mitigate drilling dysfunctions** saving millions of dollars and ultimately lowering gas prices, created an automation tool that **automatically assesses Data Quality (DQ)** and flags DQ issues across more than one million banking datasets, and helped AI companies like OpenAI and Microsoft **train their large language models (LLMs)** to produce better Python code and AI agents.

RELEVANT EXPERIENCE

Senior Developer

[Turing Enterprises](#)

📅 Jun 2025 – Dec 2025

📍 Remote

LLM Code Generation Enhancement (OpenAI / Microsoft)

- Early OpenAI models (GPT-4 and o1) frequently generated incorrect Python code, requiring targeted human feedback to improve reliability.
- Assessed complex coding scenarios and **provided rigorous RLHF feedback**, directly enabling subsequent models to produce robust, production-ready Python code.

Agentic AI Reasoning & Tool Calling (OpenAI / Microsoft)

- Early LLMs struggled with multi-step reasoning and effective tool calling necessary for complex tasks like reservations or in-depth reviews.
- **Evaluated and corrected model responses** using RLHF workflows, enabling subsequent models to successfully execute complex agentic workflows.

Manager, Data Quality and Innovations

[Bank of Nova Scotia](#)

📅 Jun 2022 – July 2024

📍 Toronto, ON

Data Quality Automation & Pipeline Architecture

- Scotiabank relied on manual processes to assess data quality (DQ) across over one million datasets, creating an urgent need for scalable automation.
- **Built PySpark data pipelines** to aggregate data from Hadoop, DB2, GCP, and local servers, **generated automated data profiles**, **implemented a Pandas-based anomaly detection** protocol that **incorporated SME feedback** to update the algorithm, and facilitate in remediation of data quality issues.
- Enabled automated data quality assessment at scale, reduced manual processing costs by **\$26 million**, increased data management productivity by **40%**, and improved data reliability for downstream AI products.

EDUCATION

Master of Data Science (MDS)

[University of British Columbia](#)

📅 2026 – 2027

📍 Vancouver, BC

TECHNICAL SKILLS

	Basic				Advanced
Python	■	■	■	■	
SQL	■	■	■	■	
R	■	■	■		
Matlab	■	■	■	■	
Airflow	■				
Docker	■				
GCP	■	■	■	■	
CLI	■				
Git	■	■	■	■	
ML	■	■	■	■	■
AI	■	■	■	■	■

LANGUAGE SKILLS

- English: Proficient
- Hindi: Native

COURSES

- Probability & Statistics
- Introduction to Python and R
- Digital Signal Processing
- System Identification
- Biostatistics

CERTIFICATES

- [Agentic AI](#)
- [LangChain for LLM Development](#)
- [Machine Learning](#)
- [Google Cloud Big Data and Machine Learning Fundamentals](#)
- [Machine Learning Engineer Certificate](#)
- [Exploring and Preparing your Data with BigQuery](#)

PATENT

- [US11499411B2](#)

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EDUCATION

Master of Science (M.Sc.)
[University of Alberta](#)
📅 2016 – 2018 📍 Edmonton, AB

Bachelor of Tech. (B.Tech.)
[Indian Institute of Technology Kanpur](#)
📅 2011 – 2016 📍 Kanpur, UP

OTHER EXPERIENCES

Senior Consultant
[EXL Services](#)
📅 2025 – 2025 📍 Gurgaon, HR
Designed and implemented automated data quality monitoring pipelines using proactive anomaly detection, significantly reducing manual oversight and enhancing operational efficiency.

iOS Developer
[Pakshi Inc.](#)
📅 2020 – 2022 📍 Calgary, AB
Engineered and launched two iOS applications, MoreToLife (social) and RoboTask (productivity), executing targeted digital marketing campaigns to drive user adoption.

Manager
[Himalaya Developers](#)
📅 2024 – 2026 📍 Rishikesh, UK
Managed end-to-end delivery of two residential projects valued at \$3.2M over an 18-month timeline, overseeing fund management, land acquisition, architectural design, contractor teams, regulatory approvals, and sales.

RESIDENCY STATUS

Permanent Resident of Canada

Research Engineer

[Pason Systems Inc.](#)

📅 Aug 2018 – Jun 2022

📍 Calgary, AB

Real-Time Drilling Dysfunction Detection & Mitigation Algorithm

- Drilling rigs needed to maximize drilling speed to reduce operational costs while protecting drill bits from costly damage.
- Designed a **real-time time-series algorithm in Java** featuring a closed-loop structure to detect and mitigate drilling dysfunctions.
- Successfully **detected 99% of dysfunction cases** and increased product adoption by **10%**.

Rotary Speed Optimization for Rate-of-Penetration (ROP)

- Drilling operations traditionally focused on weight-on-bit (WOB) as the primary control variable to increase the rate-of-penetration (ROP), leaving rotary speed underutilized.
- Worked in conjunction with **Exxon Mobil** to design a Java-based algorithm aimed at **optimizing rotary speed** to maximize ROP.
- Tested the algorithm in the field across several rigs, with results currently under study.

Predictive Rate-of-Penetration (ROP) Limiting Algorithm for Dysfunction Prevention

- Certain drilling rigs experienced frequent dysfunctions, creating a need to prevent dysfunction states proactively rather than responding after they occurred.
- Created a Java-based algorithm that **predicts upcoming system dysfunctions** and dynamically limits ROP.
- Deployed the algorithm for ongoing trials across several rigs.

In-House Drilling MATLAB Simulator

- Developing and testing new algorithms required a closed-loop system, which constrained testing capacity and slowed development.
- Developed an in-house **drilling MATLAB simulator** to test algorithms independently of QA systems availability.
- Increased testing capacity and decreased R&D time spent.

Research Assistant

[University of Alberta](#)

📅 Sep 2016 – Sep 2018

📍 Edmonton, AB

Theoretical Model for Human Postural Control

- Neurological disorders and strokes disrupt the human brain's neural signals to muscles, leaving patients unable to maintain stable sitting postures.
- Conducted experiments with healthy participants to gather physiological data, performed signal processing, and developed a theoretical mathematical model explaining how the human brain supports sitting posture.
- Established a foundational model for understanding human postural control mechanisms.